SRM TRP

TRANSPORT AI

“An AI For College Bus Check In/Out Entry”

By HARIHARAN V

I BE.EEE
SRM TRP ENGINEERING COLLEGE

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INTRODUCTION

- This document describes the vision, structure, and expected behavior of the **AI-powered Bus Entry Automation System** that I plan to develop for our college.

- The goal of this system is very straightforward: **to automatically record the entry and exit of college buses using AI, without any manual typing or delays**. The system will use CCTV cameras at both gates, detect the red circular number stickers on buses, extract the number, and instantly log the entry into an Excel sheet. Everything happens automatically in 1–2 seconds.

PURPOSE OF THE SYSTEM

- Currently, bus entry is tracked manually, which takes time and may lead to mistakes or missing entries. My idea is to replace this with a clean, automated system that: - Detects every bus when it enters or leaves the college - Reads only the **red circular number sticker** placed on the right side of buses - Logs the data instantly without human involvement - Saves logs neatly inside daily sheets, monthly files, and yearly folders - Keeps a cloud backup to avoid data loss

- This system aims to help the transport office by reducing workload, increasing accuracy, and speeding up entry during peak hours.

HOW THE SYSTEM WORKS

"The entire process is automatic. No one needs to operate the software manually once it is running".

1. CAMERA FEED

- The system continuously reads the CCTV footage from: - **Gate 1 (Main Gate)** – Already has cameras, **Gate 2 (Rear Gate)** – New camera has to be added.

2. AI DETECTION

- Whenever a bus passes through, the software looks specifically for the **red circular sticker**.

3. STICKER NUMBER READING

- The number printed inside the sticker (like 23, 41, 97, etc.) is extracted using OCR.

4. CLASSIFICATION BY TIME

- The system automatically decides whether the bus is: - **Morning Check-in** - **Afternoon Check-out Evening Check-out**

5. LOGGING

- The entry is written into an Excel/CSV sheet with details like: - Bus number - Time (AM/PM) - Gate and camera number - Entry category (morning/afternoon/evening)

6. CLOUD BACKUP

- All log files are synced to cloud storage whenever internet is available.
- This entire process takes only **1–2 seconds** per bus.

SYSTEM COMPONENTS

"To make the project easier to understand, here is a breakdown of the major parts of the system"

1. STICKER DETECTION

- The AI is trained to look only for **red circular stickers**. It ignores number plates, advertisements, bus writings, or any other numbers.

2. OCR NUMBER EXTRACTION

- This part reads the actual number printed on the sticker. Only digits are accepted.

3. DUPLICATE AVOIDANCE

- If the bus was detected just a moment ago, the system won't repeat the entry.

4. AUTOMATED TIME-BASED SORTING

- No one needs to manually divide the sheet into morning, afternoon, and evening. The software does it based on the time the bus is detected.

5. EXCEL / CSV FILE MANAGEMENT

- The system creates: - **A new sheet every day** - **A new Excel file every month** - **A new folder every year** All files are stored in an organized folder called **Bus Logs**.

6. CLOUD SYNC

- If the system is offline, logs are stored locally. Once online, everything is uploaded and synced automatically.

SOFTWARE DASHBOARD

- The software includes a simple dashboard with: - Live camera preview - Latest detections - Daily logs summary - Search bar (search by bus number) - Cloud sync status - Manual entry button (only when needed)
- It is designed to be user-friendly so that the transport office can view everything easily.

INTERNAL WORKFLOW

- Here is the path data takes inside the software:

Camera➔AI Model ➔OCR➔Time Classification➔Store➔Cloud Sync➔ Dashboard Display

Each part works together to create a smooth and reliable system.

CORE ALGORITHMS EXPLAINED

A. STICKER DETECTION ALGORITHM

1. Use YOLO-based object detection
2. Apply color mask to detect red regions
3. Apply circular-shape filter
4. Output bounding box for sticker

B. OCR PROCESSING ALGORITHM

1. Crop detected sticker
2. Convert to grayscale
3. Enhance contrast
4. Run OCR model (PaddleOCR)
5. Extract integer only (0–999)

C. DUPLICATE PREVENTION ALGORITHM

- Store last detection timestamp for each bus
- If same bus detected within X seconds ignore

D. TIME CLASSIFICATION ALGORITHM

If time is: **5 AM – 11 AM** - Morning
12 PM – 3 PM - Afternoon
3 PM – 8 PM - Evening

E. EXCEL LOGGING ALGORITHM

1. Check if monthly file exists if not, create it
2. Check if today's sheet exists if not, create it
3. Write data row
4. Save and sync to cloud

SYSTEM DIAGRAMS

A. WORKFLOW DIAGRAM

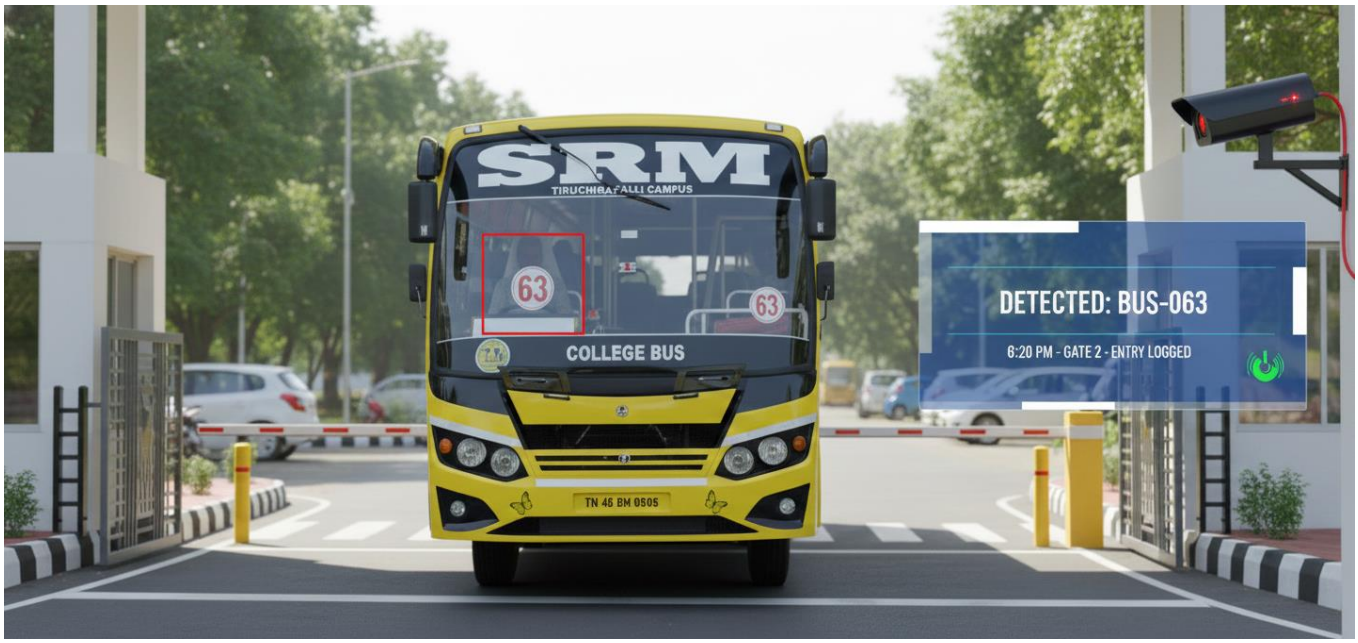
```
CCTV Camera
↓
Frame Capture
↓
AI Sticker Detection
↓
OCR Number Extraction
↓
Validation & Duplicate Removal
↓
Time Classification
↓
Excel/CSV Logging
↓
Cloud Backup
↓
Dashboard Display
```

B. ARCHITECTURE DIAGRAM

```
┌──────────────────────────┐
│      CCTV Cameras        │
└──────────────────────────┘
            │
            ↓
┌──────────────────────────┐
│  AI Detection Software   │
│  (Object Detection + OCR) │
└──────────────────────────┘
            │
            ↓
┌──────────────────────────┐
│  Data Processing Module  │
│ (Validation + Time Sort) │
└──────────────────────────┘
            │
            ↓
┌──────────────────────────┐
│   Excel Log Generator    │
└──────────────────────────┘
            │
            ↓
┌──────────────────────────┐
│     Cloud Sync Engine    │
└──────────────────────────┘
            │
            ↓
┌──────────────────────────┐
│       Dashboard UI       │
└──────────────────────────┘
```

MODEL PLACEHOLDERS

STICKER DETECTING BY CAMERA



DEMO DASHBOARD

CONCLUSION

- The **AI Bus Entry Monitoring System** successfully transforms the traditional, slow, and manual process of bus entry logging into a fully automated, accurate, and real-time digital solution. By using AI-based sticker detection, high-speed image processing, and automated Excel logging, the system ensures that bus entries are recorded within seconds, eliminating human error and reducing the workload of the transport office staff. The integration of multiple cameras, cloud backup, duplicate prevention, and time-based entry classification makes the system reliable and suitable for daily operational use.

- The dashboard interface provides clear visibility of ongoing detections, daily statistics, and complete logs, enabling the transport department to monitor and manage bus movement with ease. The modular design allows the system to grow further—supporting additional gates, more cameras, and enhanced AI capabilities as needed. Overall, this project demonstrates how AI and automation can significantly improve campus transportation management by increasing efficiency, accuracy, and safety. It lays a strong foundation for future expansions and real-world deployment within the college.

- This AI-powered bus entry system is designed to be fast, accurate, and practical for real-world use in our college. It reduces manual work, ensures clean record-keeping, and provides a modern automated approach to managing transportation.

- This document explains the system in a simple and human way, showing its purpose, workflow, benefits, and future possibilities. It serves as a strong foundation for discussions with faculty and for implementation planning.